Coexact Spectral Hydrodynamics on $L(137, 1)$: Exact Modes, Triad Couplings, and Decay

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Abstract

We study coexact spectral hydrodynamics on the round lens space $L(137, 1)$. Character formulas for spherical space forms give the signed curl gap, deck projectors, and invariant multiplicities in a fixed orientation and normalization. Specializing $SU(2)_L \times SU(2)_R$ recoupling to deck-invariant coexact modes then yields an exact $3j$–$3j$–$9j$ interaction coefficient and a fourth-root phase rule in a specified ladder-canonical basis. We give explicit nonzero witnesses and a finite two-band census, and use the coexact gap in the Hodge–Stokes energy inequality to prove exponential decay and singleton phase and trajectory attractors. These results concern one declared Hodge-viscous model; the phase is basis-dependent, and the analysis neither selects the order 137 physically nor proves global three-dimensional regularity.

1 Introduction and scope

This paper joins three mathematical layers that are often studied separately: the curl spectrum of a spherical space form, the interaction algebra of coexact vector harmonics, and the long-time energy behavior of an unforced viscous evolution. The geometry is the round quotient

$$M = L(137, 1) = S^3/\mathbb{Z}_{137}, \qquad \mathrm{Vol}(M) = \frac{2\pi^2 R_c^3}{137}.$$

The integer 137 is a mathematical specialization, not a physically selected parameter.

The spectral ingredients are specializations of established spherical-space-form character and lens-space p-form formulas [4, 19, 16, 15, 6, 9]. Likewise, the S^3 harmonic and recoupling framework is classical and appears in explicit vector-harmonic calculations [13, 3]. Helical and Beltrami decompositions and triad classifications are also classical [11, 26, 27]. Against that background, the main contribution here is an exact evaluation of the deck-restricted coexact coupling tensor on $L(p, q_0)$. The formula separates three normalization layers, identifies the complete selection algebra, and fixes a ladder-canonical phase. Level 137 is a fixed worked

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example, also used in the companion tensor-curl eta calculation; it is not selected by the model. For $L(137,1)$ we also exhibit exact witnesses and a finite tower census. The spectral gap then gives geometry-specific decay and attractor corollaries through standard energy and evolutionary-system arguments [10].

There are inequivalent viscosity operators on a Riemannian manifold [8, 23]; throughout we use the Hodge viscosity. The nonlinear three-dimensional weak evolution is therefore treated as a collection of trajectories rather than a globally single-valued solution semigroup. Section 2 fixes geometry and normalizations, Sections 4–6 derive the coupling and phase formulas and their $L(137,1)$ examples, and Sections 7–9 study decay, attractors, and the precise regularity limitation.

2 Geometry, operators, and normalizations

Write a unit quaternion as $q = x_0 + x_1\mathrm{i} + x_2\mathrm{j} + x_3\mathrm{k}$ and set $x = (x_0, x_1, x_2, x_3)$. We use the ordered imaginary basis $(e_1, e_2, e_3) = (\mathrm{i}, \mathrm{j}, \mathrm{k})$, the induced boundary orientation of $S^3 \subset \mathbb{R}^4$, and $\epsilon_{123} = +1$. The round radius-R_c metric on this unit-quaternion model is $g_{R_c} = R_c^2 g_1$. The pointwise-unit left-invariant fields

$$v_a(q) = R_c^{-1} q e_a$$

then satisfy

$$\operatorname{curl} v_a = -\frac{2}{R_c} v_a, \qquad a = 1, 2, 3.$$

Introduce complex coordinates $z_1 = x_0 + \mathrm{i}x_1$ and $z_2 = x_2 + \mathrm{i}x_3$. For $p \geq 2$ and a unit $q_0 \in (\mathbb{Z}/p\mathbb{Z})^\times$, put $\zeta = e^{2\pi\mathrm{i}/p}$ and fix the free, orientation-preserving deck generator

$$\gamma_{p,q_0}(z_1, z_2) = (\zeta z_1, \zeta^{q_0} z_2). \tag{1}$$

The quotient orientation is the one inherited from S^3. The pointwise-unit right-invariant cover frame is $w_a(q) = R_c^{-1} e_a q$ and satisfies $\operatorname{curl} w_a = +2w_a/R_c$. Our numerical space has $(p, q_0) = (137, 1)$, for which exactly $w_1(q) = R_c^{-1}\mathrm{i}q$ survives the deck action, with

$$\operatorname{curl} w_1 = +\frac{2}{R_c} w_1.$$

All four are coexact, and the Hodge Laplacian on this sector is $\Delta_H = \operatorname{curl}^2$. Thus all four have eigenvalue $4/R_c^2$.

Rank-one eta convention. In the convention of [21], the projected rank-one curl/odd-signature operator is $A_1 = *d$ on coclosed one-forms (hence on the coexact sector for these lens spaces). With the complex quotient orientation in (1), its ordinary eta invariant, not the reduced value $(\eta + h)/2$, is $\eta_1(L(p, q_0)) = -4s(q_0, p)$. The same convention takes $V_{n+2J} \otimes V_n$ as the positive branch and, on changing from unit radius to R_c, scales $A_J \mapsto R_c^{-1} A_J$ while leaving eta at zero unchanged [2, 4]. We use this citation only to align the sign, orientation, and radius conventions; all spectral, interaction, and hydrodynamic results below are derived within the present paper.

For coefficient calculations we now set $R_c = 1$. Radius is restored only in operator eigenvalues and decay rates. On complexified fields the L^2 inner product is Hermitian,

$$\langle u, v\rangle_{L^2(X)} = \int_X g_{\mathbb{C}}(u, \overline{v})\, dV.$$

Define the lifted raw trilinear by

$$T_{\text{lift,raw}}(u_1, u_2, u_3) = \int_{S^3} \det[x, u_1, u_2, u_3]\, dS.$$

Here the determinant is extended $\mathbb{C}$-multilinearly in its three field slots; no field slot is conjugated. If invariant fields u_i descend to $\bar{u}_i$ on $M = S^3/\mathbb{Z}_p$, define

$$T_{\text{quot,raw}}(\bar{u}_1, \bar{u}_2, \bar{u}_3) = p^{-1} T_{\text{lift,raw}}(u_1, u_2, u_3).$$

The cover- and quotient-normalized modes are distinct:

$$\widehat{u}^{S^3} = \frac{u}{\|u\|_{L^2(S^3)}}, \qquad \widehat{\bar{u}}^{M} = \frac{\bar{u}}{\|\bar{u}\|_{L^2(M)}};$$

the lift of $\widehat{\bar{u}}^{M}$ is $\sqrt{p}\,\widehat{u}^{S^3}$. Accordingly,

$$\begin{aligned} T_{\text{lift},L^2} &= T_{\text{lift,raw}}(\widehat{u}_1^{S^3}, \widehat{u}_2^{S^3}, \widehat{u}_3^{S^3}), \\ T_{\text{quot},L^2} &= T_{\text{quot,raw}}(\widehat{\bar{u}}_1^{M}, \widehat{\bar{u}}_2^{M}, \widehat{\bar{u}}_3^{M}) = \sqrt{p}\, T_{\text{lift},L^2}. \end{aligned}$$

These are the three normalization layers used below.

Proposition 2.1 (Gap-frame calibration)**.** *For the pointwise-unit gap frame at $p = 137$,*

$$\begin{aligned} T_{\text{lift,raw}}(v_a, v_b, v_c) &= 2\pi^2 \epsilon_{abc}, \\ T_{\text{quot,raw}}(v_a, v_b, v_c) &= \frac{2\pi^2}{137} \epsilon_{abc}, \\ T_{\text{quot},L^2}(\widehat{\bar{v}}_a^{M}, \widehat{\bar{v}}_b^{M}, \widehat{\bar{v}}_c^{M}) &= \sqrt{\frac{137}{2\pi^2}} \epsilon_{abc}. \end{aligned}$$

Every entry containing w_1 is zero. In particular, the gap tensor is totally alternating rather than a new interaction algebra.

Proof. At $R_c = 1$, the ordered ambient determinant satisfies $\det[x, v_1, v_2, v_3] = 1$ pointwise. Integration over the unit sphere gives $2\pi^2$, and determinant alternation gives the first displayed identity for all a, b, c. For a mixed component, the function $d \mapsto \int_{S^3} \det[x, v_a, v_b, w_d]\, dS$ transforms as a vector under the adjoint action of left rotations. Haar invariance makes this vector fixed by every such rotation; rotations about two independent axes force it to vanish. Alternation handles the other placements of w_d. Finally, quotient integration divides the raw value by 137, while the three quotient-normalized lifts contribute $137^{3/2}$, giving the stated L^2 layer. □

This calibration is consistent with the standard orthonormal S^3 frame and cyclic bracket conventions used in geometric hydrodynamics [22].

3 Exact coexact spectrum and deck counting

Let $E^{\mathbb{R}}_{s\ell/R_c}(X)$ denote the real coexact curl eigenspace

$$\{u \in C^\infty(TX) : \delta u^\flat = 0,\ \operatorname{curl} u = (s\ell/R_c)u\}, \qquad \ell \geq 2, \quad s \in \{-1, +1\}.$$

Magnetic and ladder bases below live in its complexification $E_{s\ell/R_c}(X) = E^{\mathbb{R}}_{s\ell/R_c}(X) \otimes_{\mathbb{R}} \mathbb{C}$. The deck-fixed bases are stable under complex conjugation, so the complex dimension of this complexification equals the real multiplicity printed below. A mode of type (ℓ, s) means a member of this complexified signed eigenspace.

For $a \geq 0$, define the deck-fixed weight count

$$F_{137}(a) = \#\{r \in \{-a, -a+2, \ldots, a\} : r \equiv 0 \pmod{137}\}.$$

The cyclotomic deck average gives the closed form

$$F_{137}(a) = \begin{cases} 2\lfloor a/274 \rfloor + 1, & a \text{ even}, \\ 2\lfloor (a+137)/274 \rfloor, & a \text{ odd}. \end{cases} \tag{2}$$

Curl-eigenfield constructions in spherical settings and lens-space quotient spectral counting are classical [9, 16, 15, 4, 19]. On bounded Euclidean balls and concentric spherical shells, Cantarella, DeTurck, Gluck, and Teytel give a complete curl-eigenfield construction under the tangent-boundary condition [7, Theorems A–B and Sections III–V]. This is adjacent spherical-domain literature, not a source for the closed lens-space multiplicities in Theorem 3.1.

Theorem 3.1 (Invariant coexact multiplicities)**.** *For a coexact level $\ell \geq 2$,*

$$\dim E_{+\ell/R_c} = F_{137}(\ell)(\ell - 1), \tag{3}$$
$$\dim E_{-\ell/R_c} = F_{137}(\ell - 2)(\ell + 1), \tag{4}$$

and both signed branches have Hodge–Stokes eigenvalue ℓ^2/R_c^2.

Proof. On the cover, the two signed coexact shells are the $SU(2)_L \times SU(2)_R$ modules

$$s = +1 :\ (j_L, j_R) = \left(\frac{\ell}{2}, \frac{\ell - 2}{2}\right), \qquad s = -1 :\ (j_L, j_R) = \left(\frac{\ell - 2}{2}, \frac{\ell}{2}\right).$$

For $L(137, 1)$ the generator (1) acts on the left magnetic factor. The cyclic character average is therefore the number of doubled left weights $r = -2j_L, -2j_L + 2, \ldots, 2j_L$ divisible by 137. Counting the positive and negative multiples of 137 in this parity lattice gives (2): an even endpoint contains zero and $2\lfloor a/274 \rfloor$ nonzero weights, whereas an odd endpoint contains the odd multiples counted by $2\lfloor (a+137)/274 \rfloor$. Multiplication by the untouched right-factor dimensions $\ell - 1$ and $\ell + 1$ proves (3) and (4). Complex conjugation preserves the fixed subspace, which identifies these complexified counts with the real dimensions. Finally, $\Delta_H = \operatorname{curl}^2$ on coexact fields, giving eigenvalue ℓ^2/R_c^2. □

The sign labels in Theorem 3.1 follow our orientation; reversing orientation exchanges them. At $\ell = 2$ the split is $1 + 3$, so the first Hodge–Stokes eigenvalue is $4/R_c^2$ with real multiplicity four. This multiplicity is specific to the exact round metric and is not asserted to persist under metric perturbation. For closed oriented three-manifolds, Kepplinger proves simplicity of the curl spectrum along generic one-parameter paths of metrics [17, Theorem 1.1]. Directly summing (3)–(4), the cumulative real coexact dimensions through the absolute-curl cutoffs $|\operatorname{curl}| \le 136/R_c, 137/R_c, 138/R_c$ are $9384, 9656, 9932$. For even $n \le 136$ the cumulative count through $|\operatorname{curl}| \le n/R_c$ is $n(n+2)/2$.

The spectrum is infinite. Its leading Weyl coefficient is $1/137$ of the cover coefficient because the quotient volume is $1/137$ of the cover volume; this is an asymptotic statement, not equality of every finite shell. At every signed cover shell the exact deck projector

$$P_{p,q_0}^{(\ell,s)} = \frac{1}{p} \sum_{k=0}^{p-1} \rho_{\ell,s}(\gamma_{p,q_0})^k$$

has image equal to the deck-invariant subspace. Here $\rho_{\ell,s}$ is the unitary pullback action of the generator (1) on $E_{s\ell/R_c}(S^3)$. Coordinates within a degenerate higher shell still require additional basis and phase choices.

For $X \in \{S^3, L(137,1)\}$ with unit curvature radius $R_c = 1$, let C_X denote the real-scalar, inhomogeneous single-constant Sobolev constant

$$C_X := \sup_{0 \neq f \in H^1(X;\mathbb{R})} \frac{\|f\|_{L^6(X)}}{\left(\|\nabla f\|_{L^2(X)}^2 + \|f\|_{L^2(X)}^2\right)^{1/2}}.$$

All norms use the unnormalized Riemannian volume of the inherited round metric. Beckner's critical sphere inequality uses normalized surface measure $d\xi = dV/(2\pi^2)$ [5, p. 4816 and p. 4818, Eq. (8)]. Converting it to unnormalized volume gives

$$\|f\|_{L^6(S^3)}^2 \le (2\pi^2)^{-2/3} \left(\|f\|_{L^2(S^3)}^2 + \frac{4}{3}\|\nabla f\|_{L^2(S^3)}^2\right).$$

The constant-function test and $4/3 > 1$ therefore give the explicit bracket

$$(2\pi^2)^{-1/3} \le C_{S^3} \le \frac{2}{\sqrt{3}}(2\pi^2)^{-1/3}. \tag{5}$$

The isometric covering lift, together with the constant-function test on the quotient of volume $2\pi^2/137$, gives

$$\left(\frac{137}{2\pi^2}\right)^{1/3} \le C_{L(137,1)} \le \frac{2}{\sqrt{3}}\left(\frac{137}{2\pi^2}\right)^{1/3}, \qquad C_{L(137,1)} \le 137^{1/3} C_{S^3}. \tag{6}$$

The ratio of the displayed upper and lower endpoints is $2/\sqrt{3}$ in both brackets. These are explicit two-sided bounds only: no endpoint is identified as the best constant, no equality case is asserted, and no sharpness claim is made inside either bracket.

4 Exact deck-restricted coexact triads

For a type (ℓ, s), as defined in Section 3, set

$$(j_L, j_R) = \begin{cases} ((\ell-2)/2, \ell/2), & s = -1, \\ (\ell/2, (\ell-2)/2), & s = +1. \end{cases}$$

The coordinate torus acts by $(z_1, z_2) \mapsto (e^{\mathrm{i}\alpha} z_1, e^{\mathrm{i}\beta} z_2)$. A polynomial vector field has weight $(w_1, w_2) \in \mathbb{Z}^2$ when its pushforward by this torus action transforms by $e^{\mathrm{i}(w_1\alpha + w_2\beta)}$. Pullback weights have the opposite signs. In the Condon–Shortley magnetic basis,

$$w_1 = m_L + m_R, \qquad w_2 = m_L - m_R, \qquad (2m_L, 2m_R) = (w_1 + w_2, w_1 - w_2),$$

where $m_L = -j_L, -j_L + 1, \ldots, j_L$ and likewise for m_R. The weights are integers for the coexact modules above. For $p \geq 2$ and $q_0 \in (\mathbb{Z}/p\mathbb{Z})^\times$, the mode descends under (1) exactly when

$$w_1 + q_0 w_2 \equiv 0 \pmod{p}. \tag{7}$$

All numerical results below specialize this free action to $(p, q_0) = (137, 1)$.

Helical/Beltrami decompositions and Fourier-space triad classifications are classical [11, 26, 27]. We specialize the corresponding recoupling machinery to deck-restricted coexact modes and derive the resulting exact selection algebra.

Call a type triple *representation-admissible* when the two Wigner rows and the three spin-one columns obey their triangle and integrality conditions; in particular $\ell_1 + \ell_2 + \ell_3$ is even. We sort types lexicographically, with $-1 < +1$. Representation admissibility is only a support condition: the $9j$ or a magnetic $3j$ may still vanish exactly.

Theorem 4.1 (Convention-complete coefficient)**.** *Let three ladder-canonical, cover-normalized modes $\widehat{u}_i^{S^3}$ form a representation-admissible triple and satisfy (7). Define*

$$\Lambda^2 = 3 \prod_{i=1}^{3} (2j_{L,i} + 1)(2j_{R,i} + 1) \begin{Bmatrix} j_{L,1} & j_{L,2} & j_{L,3} \\ 1 & 1 & 1 \\ j_{R,1} & j_{R,2} & j_{R,3} \end{Bmatrix}^2 . \tag{8}$$

If the $9j\,3j\,3j$ product is zero, then $T_{\mathrm{lift},L^2} = 0$ and no phase is assigned. Otherwise, first sort the three modes and define ϕ by Theorem 5.2; for that sorted order,

$$\begin{aligned} \pi T_{\mathrm{lift},L^2}(\widehat{u}_1^{S^3}, \widehat{u}_2^{S^3}, \widehat{u}_3^{S^3}) = \phi(\ell_i, s_i) \sqrt{\Lambda^2} \begin{pmatrix} j_{L,1} & j_{L,2} & j_{L,3} \\ m_{L,1} & m_{L,2} & m_{L,3} \end{pmatrix} \\ \times \begin{pmatrix} j_{R,1} & j_{R,2} & j_{R,3} \\ m_{R,1} & m_{R,2} & m_{R,3} \end{pmatrix} . \end{aligned} \tag{9}$$

An arbitrary ordering is defined by sorting and multiplying by the sign of the sorting permutation. Equivalently, the paired $3j$ symbols provide that odd-permutation sign while the reduced phase is permutation-invariant on the admissible domain. Thus the full trilinear is alternating, and its exact nonvanishing criterion is the nonvanishing of the displayed Wigner product.

Proof. Use Peter–Weyl matrix coefficients normalized in $L^2(S^3, dS)$ and the real Condon–Shortley spin-one column intertwiner whose highest nonzero Clebsch–Gordan coefficient is positive. Contract the three column indices with the positively oriented alternating invariant fixed by $\epsilon_{123} = +1$. The defining $9j$ recoupling identity then gives, in this real reference, the signed reduced coefficient

$$C_{\mathrm{ref}} = \sqrt{3\prod_{i=1}^{3}(2j_{L,i}+1)(2j_{R,i}+1)}\begin{Bmatrix} j_{L,1} & j_{L,2} & j_{L,3} \\ 1 & 1 & 1 \\ j_{R,1} & j_{R,2} & j_{R,3} \end{Bmatrix}.$$

The magnetic contractions left after recoupling are precisely the two Condon–Shortley $3j$ symbols in (9). Converting normalized Haar matrix coefficients to the dS-normalized fields and evaluating the remaining scalar on the oriented gap frame of Proposition 2.1 fixes the global factor to $1/\pi$; this gives $\pi T = C_{\mathrm{ref}}(3j_L)(3j_R)$. Squaring the real reduced magnitude yields (8), while Theorem 5.2 restores the exact ray and $9j$ signs. On the admissible domain $J_L + J_R = \sum_i \ell_i - 3$ is odd. Hence the two $3j$ symbols together change by the sign of an odd permutation, whereas ϕ does not, proving alternation. Deck averaging acts on each mode separately and gives (7). This is the standard S^3 recoupling mechanism [13, 3] with every normalization and row order fixed. □

Deck validity, triangle conditions, parity, and magnetic-weight sums are only necessary filters. For example,

$$\begin{pmatrix} 2 & 3 & 3 \\ 0 & -2 & 2 \end{pmatrix} = 0$$

despite coarse support. Thus "supported" never means "nonzero."

5 The ladder-canonical phase

Lemma 5.1 (Adjacent-row $9j$ sign)**.** *Let the top row be (a, b, c) and the bottom row be $(g, h, i) = (a + \sigma_1, b + \sigma_2, c + \sigma_3)$ with $\sigma_r \in \{-1, +1\}$. In a mixed shift pattern place the unique unlike shift in the middle column before taking the Racah expansion below. If k is the number of plus shifts, then every surviving term in the Racah expansion has sign $+1$ for $k = 0, 3$ and sign -1 for $k = 1, 2$. Consequently the special $9j$ symbol has that sign when its Racah support is nonempty and is structurally zero when the support is empty. The statement includes boundary rows, where one of the candidate summands may disappear.*

Proof. Column permutations have phase $+1$, since the total $9j$ spin sum is $2(a + b + c) + 2k$. Thus the unique shift of a mixed pattern may be placed in the middle column. In the exact Racah expansion

$$\sum_x (-1)^{2x}(2x+1)\begin{Bmatrix} a & b & c \\ 1 & i & x \end{Bmatrix}\begin{Bmatrix} 1 & 1 & 1 \\ b & x & h \end{Bmatrix}\begin{Bmatrix} g & h & i \\ x & a & 1 \end{Bmatrix},$$

the middle $6j$ restricts $x = b + d$. Substitution in the Racah bounds gives the complete table below. Here $S = a + b + c$, $G = g + h + i$, and (z_1, z_2, z_3) are the three Racah upper

summation integers:

$(\sigma_1,\sigma_2,\sigma_3)$	k	d	(z_1-S, z_2-2b, z_3-G)	sign
$(-,-,-)$	0	$-1, 0$	$(0,0,1),(0,1,2)$	$+,+$
$(+,+,+)$	3	$0, 1$	$(1,2,0),(2,3,0)$	$+,+$
$(-,+,-)$	1	$0, 1$	$(0,2,0),(0,3,1)$	$-,-$
$(+,-,+)$	2	$-1, 0$	$(0,0,0),(1,1,0)$	$-,-$

up to the harmless column permutations just described. Every triangle prefactor, factorial denominator, and $2x+1$ is positive; the displayed signs are the parity of $2x+z_1+z_2+z_3$ in the three single-summand $6j$ factors. Hence all surviving terms have one sign and cannot cancel. At a boundary an inadmissible value of d simply drops out, leaving the same conclusion. □

Theorem 5.2 (All-level phase rule)**.** *For a sorted representation-admissible type triple whose Wigner product in Theorem 4.1 is nonzero, put*

$$H = \frac{\ell_1+\ell_2+\ell_3}{2}, \qquad n_- = \#\{i : s_i = -1\}.$$

In the specified ladder-canonical convention, every nonzero reduced triad has

$$\phi = \mathrm{i}^{\,H+3(n_- \bmod 2)}. \tag{10}$$

Proof. In ambient coordinates $x = (x_0, x_1, x_2, x_3)$, set $z = x_0 - \mathrm{i}x_1$ and $a_\pm = x_2 \pm \mathrm{i}x_3$. The lexicographically normalized highest rays are

$$u_{\ell,+} = \mathrm{i}^{\ell-2} z^{\ell-2}(\mathrm{i}a_-, a_-, -\mathrm{i}z, -z), \qquad u_{\ell,-} = \mathrm{i}^{\ell-2} z^{\ell-2}(-\mathrm{i}a_+, -a_+, \mathrm{i}z, -z).$$

Here the component order is x_0, x_1, x_2, x_3, and monomials in each component are ordered by descending lexicographic order $x_3 > x_2 > x_1 > x_0$. The first nonzero coefficient is then $+1$. To fix the remaining ladder phases explicitly, write $w_a = e_a q$ and $v_a = q e_a$ for the ambient quaternion frames of Section 2, with $(e_1, e_2, e_3) = (\mathrm{i}, \mathrm{j}, \mathrm{k})$, and let $\mathcal{L}_X u = [X, u]$. Our lowering operators are

$$J_{L,-} = \tfrac{1}{2}\mathcal{L}_{w_2+\mathrm{i}w_3}, \qquad J_{R,-} = \tfrac{\mathrm{i}}{2}\mathcal{L}_{v_2-\mathrm{i}v_3}.$$

Starting from the displayed highest rays, normalize each lowered vector by the positive coefficient $\sqrt{(j+m)(j-m+1)}$. Thus "Condon–Shortley" includes these explicit operator phases, not only the magnitudes of the ladder coefficients.

For completeness, set $\zeta_8 = e^{\mathrm{i}\pi/4}$ and choose the coherent real Condon–Shortley reference

$$f_n = \zeta_8^n z^n, \qquad E_+ = -\mathrm{i}(\mathrm{i}a_-, a_-, -\mathrm{i}z, -z), \qquad E_- = -(-\mathrm{i}a_+, -a_+, \mathrm{i}z, -z).$$

Maximal scalar products satisfy $f_n f_m = f_{n+m}$ and all lowering coefficients are positive. Comparison with the explicit highest rays gives the per-mode phase

$$\eta_{\ell,s} = \zeta_8^{\ell+2\mathbf{1}_{\{s=-1\}}}.$$

Thus, with $k = n_-$,

$$\prod_i \eta_{\ell_i,s_i} = \zeta_8^{\sum_i \ell_i + 2k} = \mathrm{i}^{H+k}.$$

For these adjacent rows a minus-helicity mode is a plus shift, so Lemma 5.1 contributes (-1) exactly for $k = 1, 2$. Therefore

$$\phi = \mathrm{i}^{H+k}(-1)^{\mathbf{1}_{\{k=1,2\}}} = \mathrm{i}^{H+3(k \bmod 2)},$$

which is (10). If the Racah support or either magnetic $3j$ vanishes, Theorem 4.1 assigns the coefficient zero rather than an undefined phase. Finally, on the admissible domain $J_L + J_R = \sum_i \ell_i - 3$ is odd: the reduced phase is invariant under column permutation, while the paired $3j$ symbols contribute the odd-permutation sign. This is the exchange covariance required by the alternating geometric trilinear. □

Theorem 5.2 is not a physical or basis-free phase statement. Rephasing individual modes changes individual coefficient phases. It says that this fully specified basis has an exact discrete phase law.

6 Exact $L(137, 1)$ witnesses and finite tower data

This section provides exact regression examples for the coefficient formula and a finite comparison of its two specified frequency bands. The data are not an asymptotic cascade model. The actual deck restriction acts on magnetic weights, not on the level sum: the additional filter $\ell_1 + \ell_2 + \ell_3 \equiv 0 \pmod{137}$ would discard valid triads even when every $\ell_i \geq 137$. The following explicit triad proves this. At $R_c = 1$, set $w_\pm = w_2 \pm \mathrm{i}w_3$. The modes

$$u_1 = z_1^{135} w_+, \qquad u_2 = \overline{z}_1^{135} w_-$$

have curl $+137$ and weights $(2m_L, 2m_R) = (-137, -135)$ and $(137, 135)$ in this pushforward convention.

To fix the scale of the third mode, for the three prefixes used below define the zonal harmonic

$$Z_N[P] = P \sum_{k=0}^{N} c_k |z_1|^{2k} |z_2|^{2(N-k)}$$

as the unique harmonic polynomial of this form normalized by $c_N = 1$. Put

$$f_0 = Z_{68}[1], \qquad f_+ = Z_{67}[z_1 z_2], \qquad f_- = Z_{67}[\overline{z}_1\, \overline{z}_2]$$

and choose the concrete weight-$(0, 0)$ curl-$+138$ representative

$$u_3 = -f_+ w_- + \frac{\mathrm{i}}{34} f_0 w_1 + f_- w_+. \tag{11}$$

Thus no free ray scale remains. All three modes descend to $L(137, 1)$, and exact integration gives

$$T_{\mathrm{lift,raw}}(u_1, u_2, u_3) = \frac{187232862460210808175}{10632982165841828043216257332726568990 38}\pi^2 \neq 0.$$

Since $137 + 137 + 138 = 412 \equiv 1 \pmod{137}$, the congruence is not necessary. The exact criterion remains the Wigner product in Theorem 4.1; this example makes no separate statement about sufficiency.

For a direct signed tower anchor take

$$((\ell,s)_1,(\ell,s)_2,(\ell,s)_3)=((137,+),(139,-),(274,+)),$$

with

$$2m_L=(137,137,-274),\qquad 2m_R=(135,137,-272).$$

The raw magnitude below is attached to concrete representatives, rather than to three freely rescalable rays. Let

$$J_{R,-}=J_{R,1}-\mathrm{i}J_{R,2},\qquad J_{R,-}|j,m\rangle=\sqrt{(j+m)(j-m+1)}\,|j,m-1\rangle$$

be the Condon–Shortley-normalized right lowering operator, and set

$$U_1=u_{137,+},\qquad U_2=J_{R,-}u_{139,-},\qquad U_3=\overline{u_{274,+}}.$$

These representatives have exactly the displayed magnetic weights. If $\|U_i\|^2_{L^2(S^3)}=\pi^2N_i$, direct exact integration gives $N_1N_2N_3=556/320229$. The lifted, cover-unnormalized representative integral has magnitude

$$|T_{\mathrm{lift,raw}}(U_1,U_2,U_3)|=\frac{180}{12467}\pi^2,$$

independently of the phase chosen for the ambient right lowering operator. The cover-normalized lifted square is

$$\pi^2\left|T_{\mathrm{lift},L^2}(\widehat{U}_1^{S^3},\widehat{U}_2^{S^3},\widehat{U}_3^{S^3})\right|^2=\frac{28503900}{237409081}.$$

They are related, not identical:

$$\left(\frac{180}{12467}\right)^2=\frac{28503900}{237409081}\frac{556}{320229}.\tag{12}$$

The finite window

$$\{137,138,139,140\}\cup\{274,275,276,277\}$$

contains both helicities. A reduced key consists of a lexicographically sorted type triple together with a doubled-left-weight triple $c=(2m_{L,1},2m_{L,2},2m_{L,3})$. Each c_i lies in its spin lattice, is divisible by 137, and satisfies $c_1+c_2+c_3=0$. For repeated adjacent types we impose $c_1\geq c_2$ or $c_2\geq c_3$, respectively, to quotient only the corresponding exchange duplication. Put

$$Q(e)=\Lambda(e)^2\begin{pmatrix}j_{L,1}&j_{L,2}&j_{L,3}\\ m_{L,1}&m_{L,2}&m_{L,3}\end{pmatrix}^2.$$

The reduced amplitude is total on every stored key, but no phase is assigned to a structural zero:

$$A_{\mathrm{red}}(e)=\begin{cases}0, & Q(e)=0,\\ \phi(e)\sqrt{\Lambda(e)^2}\begin{pmatrix}j_{L,1}&j_{L,2}&j_{L,3}\\ m_{L,1}&m_{L,2}&m_{L,3}\end{pmatrix}, & Q(e)>0,\end{cases}\qquad R(e)=|A_{\mathrm{red}}(e)|^2.$$

Thus $R(e) = \Lambda(e)^2(3j_L(e))^2$, while the right magnetic dependence remains factored: $\pi T_{\mathrm{lift},L^2} = A_{\mathrm{red}}(e)(3j_R)$.

There are 816 sorted type triples in the window. Of these, 171 cross-band types and 58 all-high-band types have a nonzero reduced key. Let $\mathcal{B}_X$ and $\mathcal{B}_{H2}$ be their respective nonzero reduced key sets; they contain 385 and 160 entries. The inherited first-band block $\mathcal{B}_0$ stores 56 reduced keys, 46 with nonzero A_{red}. Expanding the right-weight $3j_R$ factor gives 806,408 eligible indexed completion slots across all 56 stored keys, including keys whose reduced factor vanishes. It is therefore neither a right-resolved enumeration nor a count of nonzero pointwise coefficients.

For $B \in \{0, X, H2\}$ define the coherent reduced sum

$$S_B = \sum_{e \in \mathcal{B}_B} A_{\mathrm{red}}(e) = \sum_{(r,a,b) \in \mathcal{L}_B} (a + \mathrm{i}b)\sqrt{r}.$$

Equal positive rational radicands are aggregated, and zero aggregate coefficients are removed. The exact-arithmetic supplement gives the complete canonical lists $\mathcal{L}_0$, $\mathcal{L}_X$, and $\mathcal{L}_{H2}$, with 28, 149, and 73 rows, respectively, together with directed rational enclosures for every sum. Numerically,

$$\begin{aligned} S_0 &\approx -0.215211944909057 + 1.72881536303333 \times 10^{-7}\mathrm{i}, \\ S_X &\approx 40.3109087370940 + 0.444138023269106\mathrm{i}, \\ S_{H2} &\approx 15.4464538323349 - 0.0742113043164842\mathrm{i}. \end{aligned}$$

Define

$$M_{H1} = \max_{e \in \mathcal{B}_0} R(e), \qquad M_{H2} = \max_{e \in \mathcal{B}_{H2}} R(e).$$

With the canonical key ordering, the first-band maximizer is

$$((138, -), (140, -), (140, -)), \qquad 2m_L = (0, 0, 0).$$

The two sign-related high-band maximizers have

$$((276, -), (277, -), (277, -)), \qquad 2m_L = (-274, 137, 137) \text{ or } (274, -137, -137).$$

The exact statistic $M_{H2}/M_{H1} = N/D$ is recorded in the supplement. Its decimal begins 20.508403, and the corresponding reduced-amplitude ratio is

$$\sqrt{M_{H2}/M_{H1}} \approx 4.52862051343485.$$

All tower counts, ratios, and cancellations in this section are finite-window, convention-dependent data. They imply no asymptotic growth, monotonicity, cascade, turbulence, or typicality law.

7 Unforced Hodge–Stokes decay

The Leray–Hopf weak framework and energy–Poincaré decay mechanism are classical [20, 14, 25, 10]. Here we specialize them to the declared Hodge operator and its exact quotient gap.

We now fix the viscous model. Let

$$H = \overline{\{v \in C^\infty(TM) : \operatorname{div} v = 0\}}^{L^2}, \qquad \mathbb{P} : L^2(TM) \longrightarrow H$$

be the real solenoidal phase space and Leray projector. Musical identifications are understood. With the positive-sign convention $\Delta_H = d\delta + \delta d$, define

$$A = \mathbb{P}\Delta_H, \qquad D(A) = H^2(TM) \cap H, \qquad V = D(A^{1/2}).$$

Since $H^1(M;\mathbb{R}) = 0$, this is the positive Hodge–Stokes operator on the coexact sector. We study only

$$\partial_t u + \mathbb{P}(\nabla_u u) + \nu A u = 0, \qquad u(0) = u_0 \in H, \qquad \nu > 0. \tag{13}$$

No conclusion below is transferred to a rough-Laplacian or deformation-tensor viscosity.

An admissible Leray–Hopf trajectory is a distributional solution

$$u \in C_{\mathrm{w}}([0,\infty); H) \cap L^\infty_{\mathrm{loc}}([0,\infty); H) \cap L^2_{\mathrm{loc}}([0,\infty); V)$$

for which there is a full-measure set of admissible starting times $\mathcal{G}_u \subset [0,\infty)$, containing the prescribed initial time 0, such that

$$\|u(t)\|_H^2 + 2\nu \int_s^t \|A^{1/2}u(r)\|_H^2 \, dr \le \|u(s)\|_H^2, \qquad s \in \mathcal{G}_u, \quad t \ge s. \tag{14}$$

For complete trajectories the same definition is used on $\mathbb{R}$, with $\mathcal{G}_u \subset \mathbb{R}$ of full measure. For each datum in H, such forward trajectories are obtained by the standard solenoidal Galerkin construction [20, 14, 25]: project onto the coexact eigenfields, use the cancellation $\langle \nabla_u u, u\rangle = 0$, and pass to a subsequence using compact $H^1 \hookrightarrow L^2$ on the closed quotient. The energy bound and weak continuity give the stated admissible-time inequality. This supplies existence, not uniqueness or global strong regularity.

The coexact Poincaré inequality is

$$\|A^{1/2}v\|_H^2 \ge \lambda_1 \|v\|_H^2, \qquad \lambda_1 = 4/R_c^2.$$

Combining this inequality with (14), applying the integral Gronwall argument at admissible starting times, and using weak continuity and lower semicontinuity at terminal times gives, for every $s \in \mathcal{G}_u$ and $t \ge s$,

$$\|u(t)\|_H^2 \le \|u(s)\|_H^2 e^{-2\nu\lambda_1(t-s)} = \|u(s)\|_H^2 \exp\left[-\frac{8\nu(t-s)}{R_c^2}\right]. \tag{15}$$

Hence

$$\|u(t)\|_H \le \|u(s)\|_H \exp\left[-\frac{4\nu(t-s)}{R_c^2}\right]. \tag{16}$$

8 Evolutionary and trajectory attractors

We use standard dissipative and evolutionary-system language [25, 10], while retaining set-valued dynamics because uniqueness is unavailable.

Let $\mathcal{E}$ be the collection of all admissible Leray–Hopf trajectories. For bounded $B \subset H$, define the states reachable at time t by

$$\mathcal{U}(t)B = \{u(t) : u \in \mathcal{E},\ u(0) \in B\}.$$

This notation is set-valued; no semigroup or composition law is asserted. For $X, Y \subset H$, let

$$\operatorname{dist}_H(X, Y) = \sup_{x \in X} \inf_{y \in Y} \|x - y\|_H$$

be the one-sided Hausdorff semidistance. If $R_B = \sup_{b \in B} \|b\|_H$, then

$$\operatorname{dist}_H(\mathcal{U}(t)B, \{0\}) \le R_B e^{-4\nu t/R_c^2}. \tag{17}$$

Let $\mathcal{K}^+$ be the forward trajectories, endowed with the local weak topology inherited from $C_{\mathrm{loc}}([0,\infty); H_{\mathrm{w}})$. Define $(\tau_h u)(r) = u(r+h)$ for $h \ge 0$. Let $\mathcal{K} \subset C_{\mathrm{loc}}(\mathbb{R}; H_{\mathrm{w}})$ be the complete trajectories bounded in H, with the analogous translation action.

Theorem 8.1 (Zero phase and trajectory attractors)**.** *The strong $H = L^2$ phase attractor is $\{0\}$. Moreover, $\mathcal{K}$ consists only of the constant-zero complete trajectory, and its restriction to $[0,\infty)$ is the trajectory attractor in the local weak topology.*

Proof. Equation (17) proves uniform strong attraction of every bounded phase set. Zero initial energy forces the zero trajectory by (14), so the singleton is invariant; every closed attracting set must contain it.

If $u \in \mathcal{K}$ and $M = \sup_{r \in \mathbb{R}} \|u(r)\|_H$, then for fixed t choose $s_n \in \mathcal{G}_u$ with $s_n < t$ and $s_n \to -\infty$. Equation (15) gives

$$\|u(t)\|_H^2 \le M^2 e^{-8\nu(t-s_n)/R_c^2} \longrightarrow 0.$$

Thus $u(t) = 0$. Finally, for every $T > 0$, trajectories whose initial values have H-radius at most R satisfy

$$\sup_{0 \le r \le T} \|(\tau_t u)(r)\|_H \le R e^{-4\nu t/R_c^2}.$$

The positive restriction of the zero complete trajectory therefore attracts in $C_{\mathrm{loc}}([0,\infty); H_{\mathrm{w}})$, in fact with the displayed strong bound. Its dimension is zero because it is a singleton, not because of a spectral cutoff. □

9 Regularity boundary and conditional continuation

The energy-class interpolation is the standard Ladyzhenskaya bound, while the regularity comparison is the classical Serrin criterion [18, 24]. Use the Hodge graph norm

$$\|v\|_{H^1_A}^2 = \|v\|_H^2 + \|A^{1/2} v\|_H^2, \qquad v \in V.$$

At unit curvature radius, the positive-sign Weitzenböck convention is $\Delta_H = \nabla^*\nabla + \mathrm{Ric}$ with $\mathrm{Ric} = 2g$ on one-forms [1, Section 2.2, Theorems 2.6–2.7]. Hence, for every smooth coexact one-form v,

$$\|A^{1/2}v\|_2^2 = \|\nabla v\|_2^2 + 2\|v\|_2^2.$$

The quadratic-form Kato inequality [12, Corollary 2.5] then gives

$$\|\nabla|v|\|_2^2 + \|v\|_2^2 \le \|\nabla v\|_2^2 + \|v\|_2^2 \le \|v\|_{H^1_A}^2.$$

For $v \in V$, choose smooth coexact $v_n \to v$ in the graph norm. Applying the closed Weitzenböck form identity to the graph-norm convergence gives $v_n \to v$ in L^2 and $\nabla v_n \to \nabla v$ in L^2. Kato bounds $|v_n|$ in scalar H^1, while $\big||v_n| - |v|\big| \le |v_n - v|$ gives strong L^2 convergence to $|v|$. A subsequence converges weakly in H^1 to that same limit. Weak lower semicontinuity, followed by $\|\nabla v_n\|_2 \to \|\nabla v\|_2$, gives $\|\nabla|v|\|_2 \le \|\nabla v\|_2$. The scalar Sobolev inequality therefore passes to the graph-form domain, giving

$$C_V := \frac{2}{\sqrt{3}}\left(\frac{137}{2\pi^2}\right)^{1/3}, \qquad \|v\|_{L^6(L(137,1))} \le C_{L(137,1)}\|v\|_{H^1_A} \le C_V\|v\|_{H^1_A} \qquad (R_c = 1). \tag{18}$$

This graph-form embedding is one-way: small L^6 norm does not imply small graph norm, so an independently supplied graph-norm threshold δ cannot be replaced by δ/C_V as an L^6 threshold. The energy class and the Sobolev embedding $V \hookrightarrow L^6(M)$ give, for $0 \le \theta \le 1$,

$$u \in L^p_{\mathrm{loc}}(L^q), \qquad \frac{1}{p} = \frac{\theta}{2}, \qquad \frac{1}{q} = \frac{1-\theta}{2} + \frac{\theta}{6}. \tag{19}$$

Thus $2 \le p \le \infty$ and $2 \le q \le 6$, with the exponents coupled by

$$\frac{2}{p} + \frac{3}{q} = \frac{3}{2},$$

not the Serrin region $2/p + 3/q \le 1$ (with $q > 3$ in its usual nonendpoint form). In particular, $(p, q) = (\infty, 6)$ is not supplied by (19): $p = \infty$ corresponds to $q = 2$. The exact spectrum is infinite, so finite cutoff counts do not imply cascade starvation.

For the exact time-average statement this graph norm is equivalent to the usual H^1 norm on the solenoidal sector, but the constant below is asserted only for this declared graph norm. Put $E(t) = \|u(t)\|_H^2$ and $E_0 = E(0)$, with no factor $1/2$. For every $t \in \mathcal{G}_u$ and $\Delta t > 0$, the energy inequality gives an admissible $s \in \mathcal{G}_u \cap [t, t + \Delta t]$ such that

$$\|u(s)\|_{H^1_A}^2 \le E(t)\left(1 + \frac{1}{2\nu\Delta t}\right). \tag{20}$$

The regularity consequence is explicitly conditional. Assume an independent theorem supplies $\delta > 0$ for equation (13), the given (ν, R_c), and this exact graph norm, such that every $v \in V$ with $\|v\|_{H^1_A} \le \delta$ generates a global strong continuation, and that weak–strong uniqueness identifies any Leray–Hopf trajectory having datum v with that continuation. Then the original trajectory is strong from the selected s onward whenever the right-hand side of (20) is at most δ^2. Since $E(t) \le E_0 e^{-2\nu\lambda_1 t}$, this holds for every admissible t satisfying

$$t \ge \frac{1}{2\nu\lambda_1}\log\left[\frac{E_0}{\delta^2}\left(1 + \frac{1}{2\nu\Delta t}\right)\right], \qquad \lambda_1 = \frac{4}{R_c^2}. \tag{21}$$

The conditional onset is then no later than $t + \Delta t$. Because $\mathcal{G}_u$ has full measure, admissible t may be chosen arbitrarily close above the displayed threshold. This is neither an unconditional T^* formula nor a theorem of global three-dimensional regularity or uniqueness.

10 Conclusion

We have combined the exact coexact spectrum of $L(137, 1)$ with its deck-restricted interaction algebra. The result includes a convention-complete Wigner formula, a discrete ladder-canonical phase, explicit nonzero witnesses, and finite two-band data. The same spectral gap yields exponential energy decay and singleton phase and trajectory attractors for the declared Hodge–Stokes evolution, without assuming uniqueness of weak solutions.

Several questions remain open. The finite tower census does not determine an asymptotic cascade law, the canonical phase is not invariant under arbitrary rephasing, and the conditional continuation argument does not prove global three-dimensional regularity. It would be particularly useful to understand how much of the interaction algebra survives beyond round quotients and whether the exact selection rules constrain long-range transfer across the full, infinite spectrum.

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References

[1] Masayuki Aino. Lichnerowicz–Obata estimate, almost parallel p-form and almost product manifolds. *Journal of Geometric Analysis*, 31(11):10915–11001, 2021.

[2] Michael F. Atiyah, Vijay K. Patodi, and Isadore M. Singer. Spectral asymmetry and riemannian geometry. II. *Mathematical Proceedings of the Cambridge Philosophical Society*, 78(3):405–432, 1975.

[3] Stefano Baiguera, Troels Harmark, Yang Lei, and Nico Wintergerst. Symmetry structure of the interactions in near-BPS corners of $\mathcal{N} = 4$ super-Yang–Mills. *Journal of High Energy Physics*, 2021(4):029, 2021.

[4] Christian Bär. The curl operator on odd-dimensional manifolds. *Journal of Mathematical Physics*, 60(3):031501, 2019.

[5] William Beckner. Sobolev inequalities, the Poisson semigroup, and analysis on the sphere S^n. *Proceedings of the National Academy of Sciences of the United States of America*, 89(11):4816–4819, 1992.

[6] J. Ben Achour, E. Huguet, J. Queva, and J. Renaud. Explicit vector spherical harmonics on the 3-sphere. *Journal of Mathematical Physics*, 57(2):023504, 2016.

[7] Jason Cantarella, Dennis DeTurck, Herman Gluck, and Mikhail Teytel. The spectrum of the curl operator on spherically symmetric domains. *Physics of Plasmas*, 7(7):2766–2775, 2000.

[8] Chi Hin Chan, Magdalena Czubak, and Marcelo M. Disconzi. The formulation of the Navier–Stokes equations on Riemannian manifolds. *Journal of Geometry and Physics*, 121:335–346, 2017.

[9] Subrahmanyan Chandrasekhar and Paul C. Kendall. On force-free magnetic fields. *The Astrophysical Journal*, 126(2):457–460, 1957.

[10] Alexey Cheskidov and Ciprian Foias. On global attractors of the 3d Navier–Stokes equations. *Journal of Differential Equations*, 231(2):714–754, 2006.

[11] Peter Constantin and Andrew Majda. The Beltrami spectrum for incompressible fluid flows. *Communications in Mathematical Physics*, 115(3):435–456, 1988.

[12] Batu Güneysu. Kato's inequality and form boundedness of Kato potentials on arbitrary Riemannian manifolds. *Proceedings of the American Mathematical Society*, 142(4):1289–1300, 2014.

[13] Ken-ji Hamada and Shinichi Horata. Conformal algebra and physical states in a non-critical 3-brane on $R \times S^3$. *Progress of Theoretical Physics*, 110:1169–1210, 2003.

[14] Eberhard Hopf. über die anfangswertaufgabe für die hydrodynamischen grundgleichungen. *Mathematische Nachrichten*, 4(1–6):213–231, 1950.

[15] Akira Ikeda. On lens spaces which are isospectral but not isometric. *Annales scientifiques de l'École Normale Supérieure*, 13(3):303–315, 1980.

[16] Akira Ikeda and Yoshihiko Yamamoto. On the spectra of 3-dimensional lens spaces. *Osaka Journal of Mathematics*, 16(2):447–469, 1979.

[17] Willi Kepplinger. On spectral simplicity of the Hodge laplacian and curl operator along paths of metrics. *Transactions of the American Mathematical Society*, 377(11):7829–7845, 2024.

[18] Olga A. Ladyzhenskaya. *The Mathematical Theory of Viscous Incompressible Flow*. Gordon and Breach, New York, 2 edition, 1969.

[19] Emilio A. Lauret. The spectrum on p-forms of a lens space. *Geometriae Dedicata*, 197:107–122, 2018.

[20] Jean Leray. Sur le mouvement d'un liquide visqueux emplissant l'espace. *Acta Mathematica*, 63:193–248, 1934.

[21] Luqman Mahmood. Eta Invariants of Transverse-Traceless Tensor Curl on Three-Dimensional Lens Spaces. Companion preprint. https://doi.org/10.5281/zenodo.21433675, 2026.

[22] Klas Modin and Stephen C. Preston. Zeitlin's model for axisymmetric 3d Euler equations. *Nonlinearity*, 38(2):025008, 2025.

[23] Maryam Samavaki and Jukka Tuomela. Navier–Stokes equations on Riemannian manifolds. *Journal of Geometry and Physics*, 148:103543, 2020.

[24] James Serrin. On the interior regularity of weak solutions of the Navier–Stokes equations. *Archive for Rational Mechanics and Analysis*, 9(1):187–195, 1962.

[25] Roger Temam. *Infinite-Dimensional Dynamical Systems in Mechanics and Physics*, volume 68 of *Applied Mathematical Sciences*. Springer, New York, 2 edition, 1997.

[26] Fabian Waleffe. The nature of triad interactions in homogeneous turbulence. *Physics of Fluids A*, 4(2):350–363, 1992.

[27] Fabian Waleffe. Inertial transfers in the helical decomposition. *Physics of Fluids A*, 5(3):677–685, 1993.